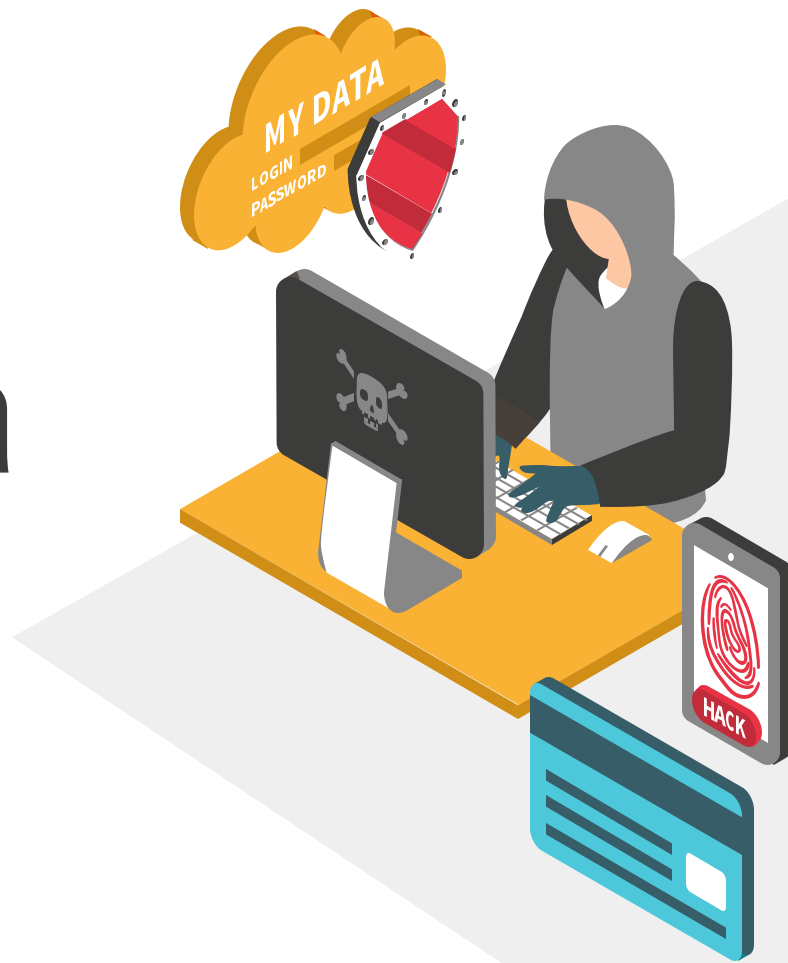


SURICATA **(DPI and Lua Scripting)**

Licenciatura em Engenharia Informática
U.C. Segurança
Artur Falcão - 2024130485
Gabriel Rodrigues - 2024133397
Rodrigo Prazeres - 2023139373



O que é o Suricata?

- Ferramenta de segurança de redes;
- Pode funcionar como IDS ou IPS;
- Analisa tráfego em tempo real;
- Deteta atividades suspeitas;
- Gera alertas de possíveis ataques.



Figura 1 - Suricata logótipo [1].

DPI: Deep Packet Inspection

- Atua na camada de aplicação
- Deteta payloads suspeitos
- Base para os scripts Lua

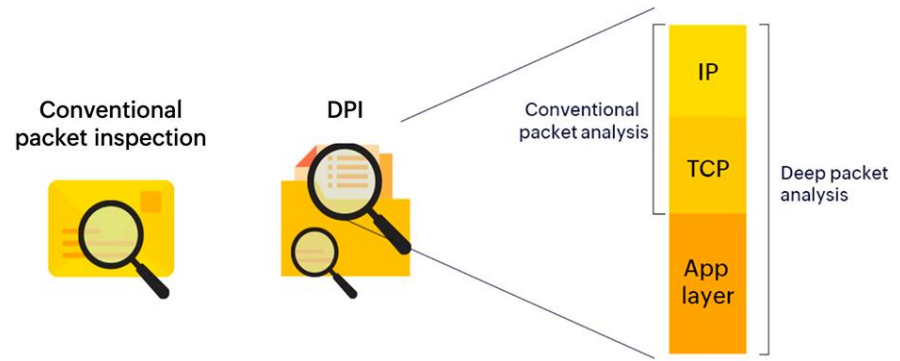


Figura 2 - Esquema DPI.

Lua *scripts*

- Scripts .lua são códigos com lógica personalizada;
- Usados para complementar as regras do Suricata;
- Tornam a deteção mais flexível e adaptável.



Figura 3 - Lua logótipo [2].

Como ligar o Lua ao Suricata?

- Regras definidas no suricata.rules;
- Cada regra indica o tráfego a analisar;
- As regras podem chamar scripts Lua;
- O Lua aplica lógica personalizada.

```
GNU nano 6.2 /etc/suricata/rules/suricata.rules *
alert http any any -> any any (msg:"[ALERTA LUA] Fuga de Cartao de Credito Detetada"; lua:/etc/suricata/lua/dlp_cartao.lua; sid:100001; rev:1;)
alert tcp any any -> any any (msg:"[ALERTA LUA] Ataque de Forca Bruta Detetado"; lua:/etc/suricata/lua/forca_bruta_contador.lua; sid:100002; rev:1;)
alert tcp any any -> any any (msg:"[ALERTA LUA] Tool de Ataque Detetada"; lua:/etc/suricata/lua/detecao_agente.lua; sid:100003; rev:1;)
alert tcp any any -> any any (msg:"[ALERTA LUA] Download Executavel Detetado"; lua:/etc/suricata/lua/bloq_exec.lua; sid:100004; rev:1;)
```

Figura 4 - Exemplo de regras no suricata.rules usadas para chamar os scripts Lua da demonstração.

A nossa test bed no GNS3

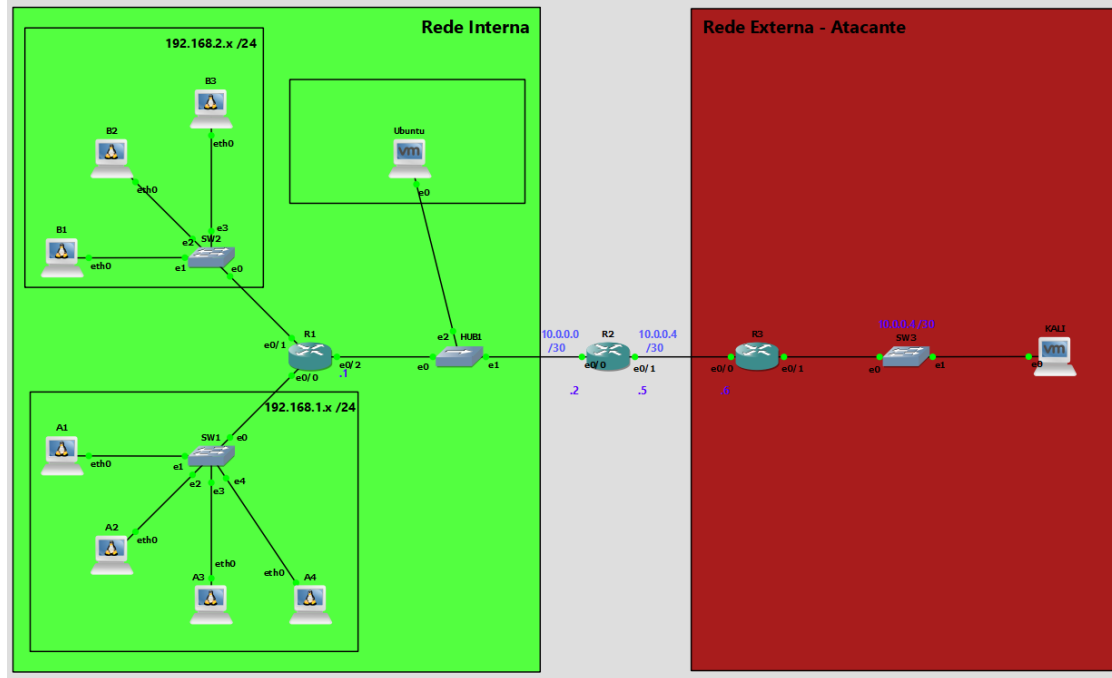


Figura 5 - Topologia.

Experiências e Análise de Deteções



DLP

Deteta possíveis fugas de dados de cartões de crédito.



User-Agent

Identifica ferramentas como sqlmap, nmap e nikto.



Força Bruta

Conta tentativas falhadas e gera alerta após 3 erros.



Block Exec

Alerta de ficheiros .exe, .sh e .bat.



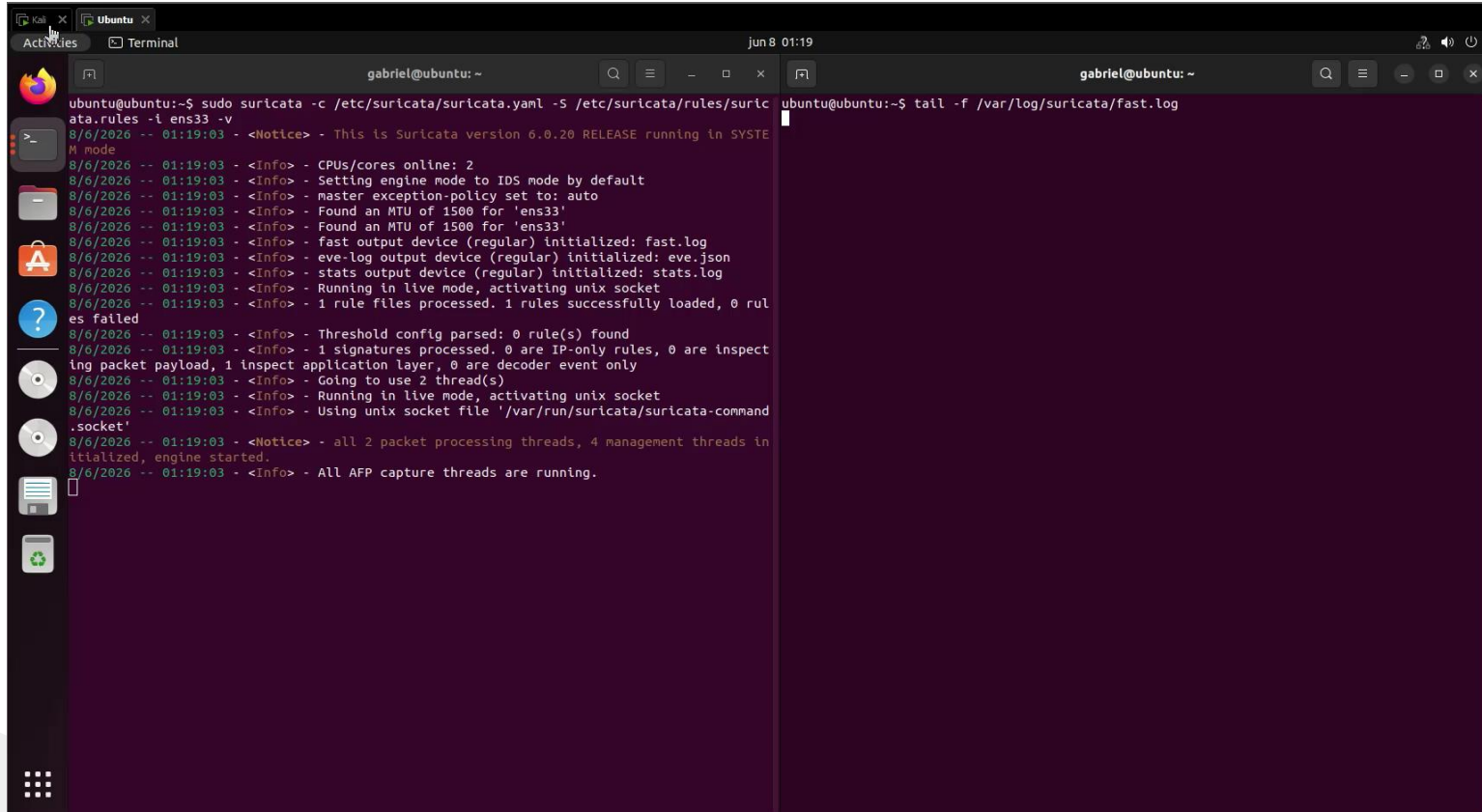
**Instituto Superior
de Engenharia**
Politécnico de Coimbra

Demonstração



GNS3®

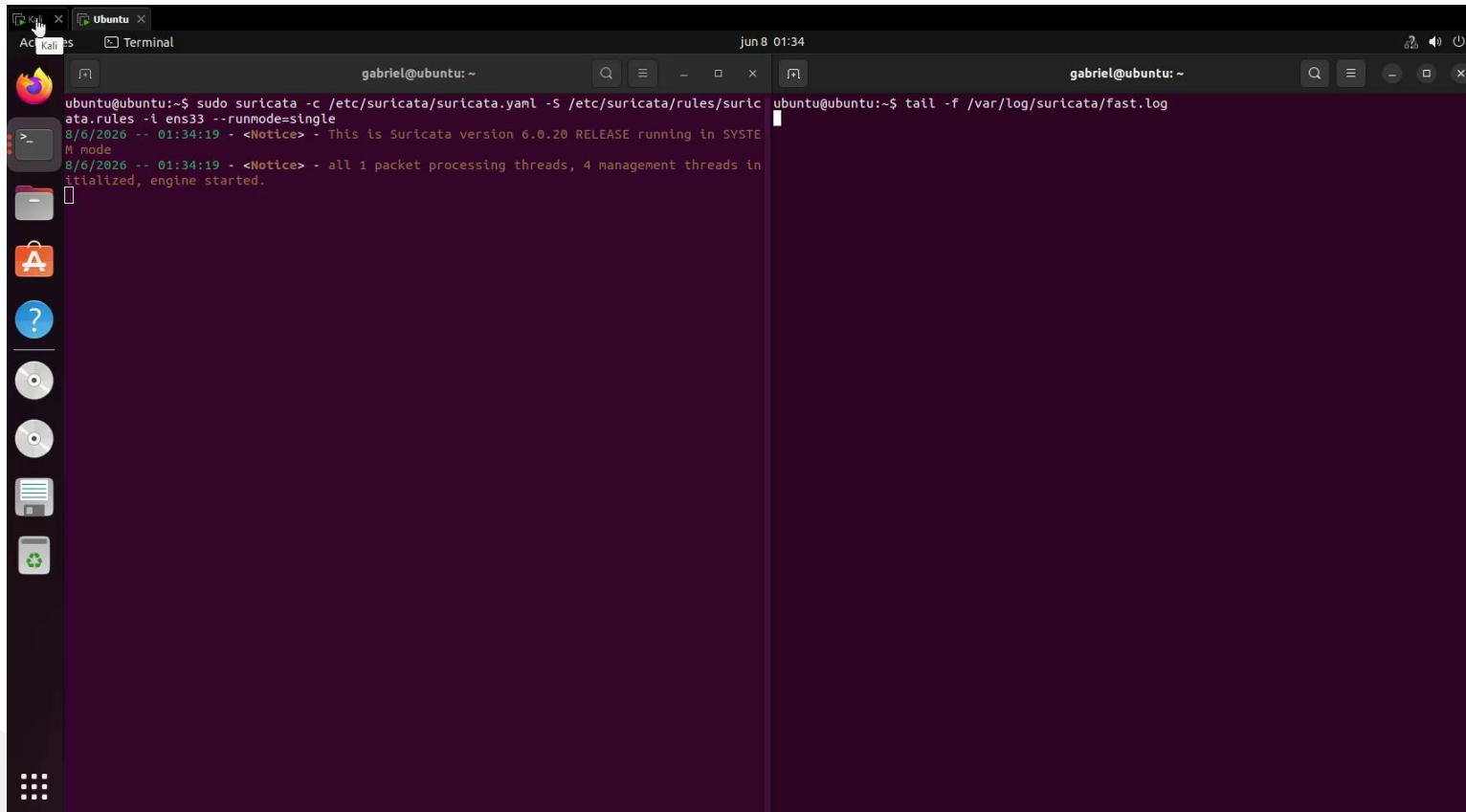
DLP - Data Loss Prevention



```
ubuntu@ubuntu:~$ sudo suricata -c /etc/suricata/suricata.yaml -S /etc/suricata/rules/suricata.rules -l ens33 -v
8/6/2026 -- 01:19:03 - <Notice> - This is Suricata version 6.0.20 RELEASE running in SYST
M mode
8/6/2026 -- 01:19:03 - <Info> - CPUs/cores online: 2
8/6/2026 -- 01:19:03 - <Info> - Setting engine mode to IDS mode by default
8/6/2026 -- 01:19:03 - <Info> - master exception-policy set to: auto
8/6/2026 -- 01:19:03 - <Info> - Found an MTU of 1500 for 'ens33'
8/6/2026 -- 01:19:03 - <Info> - Found an MTU of 1500 for 'ens33'
8/6/2026 -- 01:19:03 - <Info> - fast output device (regular) initialized: fast.log
8/6/2026 -- 01:19:03 - <Info> - eve-log output device (regular) initialized: eve.json
8/6/2026 -- 01:19:03 - <Info> - stats output device (regular) initialized: stats.log
8/6/2026 -- 01:19:03 - <Info> - Running in live mode, activating unix socket
8/6/2026 -- 01:19:03 - <Info> - 1 rule files processed. 1 rules successfully loaded, 0 rul
es failed
8/6/2026 -- 01:19:03 - <Info> - Threshold config parsed: 0 rule(s) found
8/6/2026 -- 01:19:03 - <Info> - 1 signatures processed. 0 are IP-only rules, 0 are inspect
ing packet payload, 1 inspect application layer, 0 are decoder event only
8/6/2026 -- 01:19:03 - <Info> - Going to use 2 thread(s)
8/6/2026 -- 01:19:03 - <Info> - Running in live mode, activating unix socket
8/6/2026 -- 01:19:03 - <Info> - Using unix socket file '/var/run/suricata/suricata-command
.socket'
8/6/2026 -- 01:19:03 - <Notice> - all 2 packet processing threads, 4 management threads in
itIALIZED, engine started.
8/6/2026 -- 01:19:03 - <Info> - All AFP capture threads are running.

ubuntu@ubuntu:~$ tail -f /var/log/suricata/fast.log
```

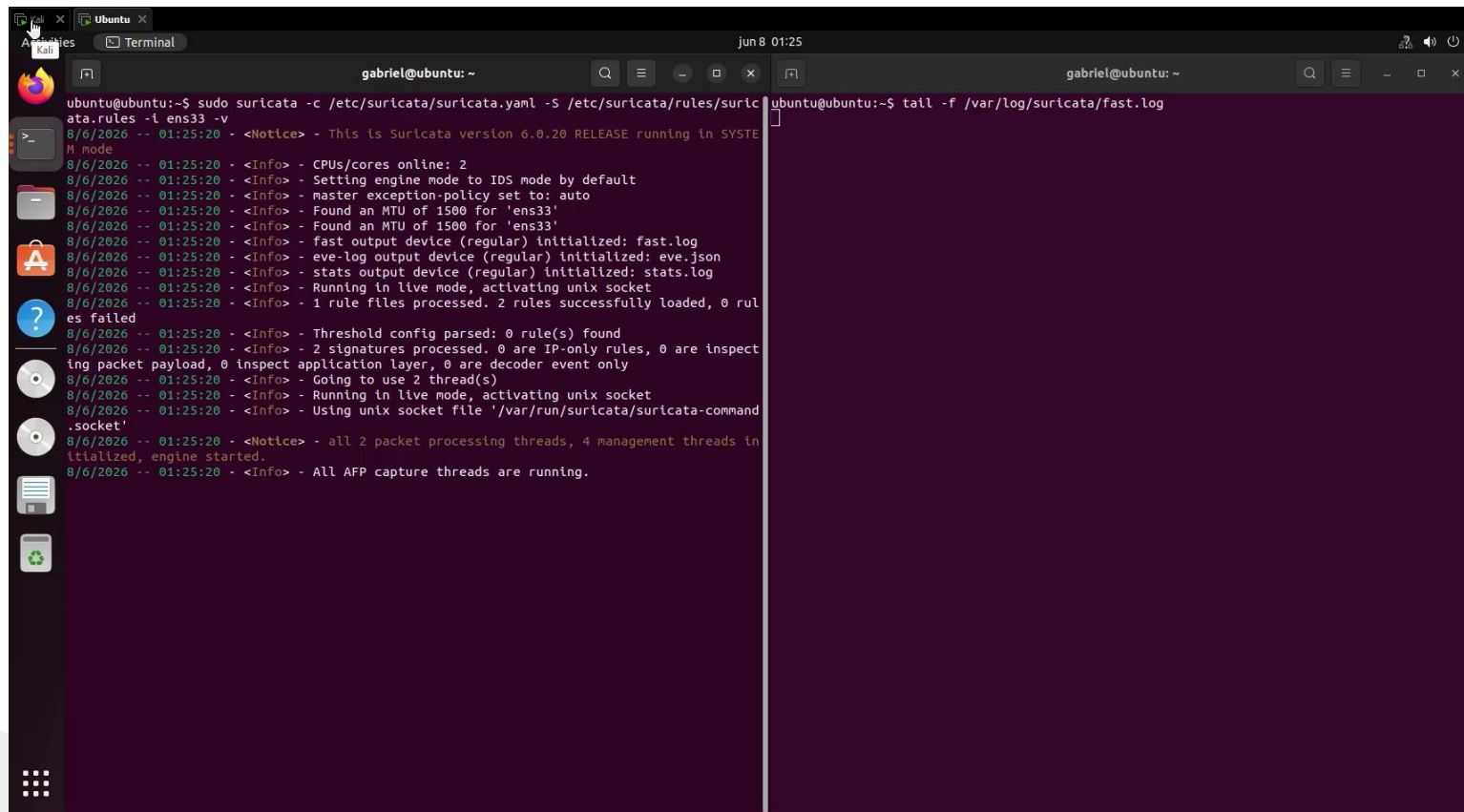
Força Bruta



```
ubuntu@ubuntu:~$ sudo suricata -c /etc/suricata/suricata.yaml -S /etc/suricata/rules/suricata.rules -i ens33 --runmode=single
8/6/2026 -- 01:34:19 - <Notice> - This is Suricata version 6.0.20 RELEASE running in SYSTEM mode
8/6/2026 -- 01:34:19 - <Notice> - all 1 packet processing threads, 4 management threads initialized, engine started.

ubuntu@ubuntu:~$ tail -f /var/log/suricata/fast.log
```

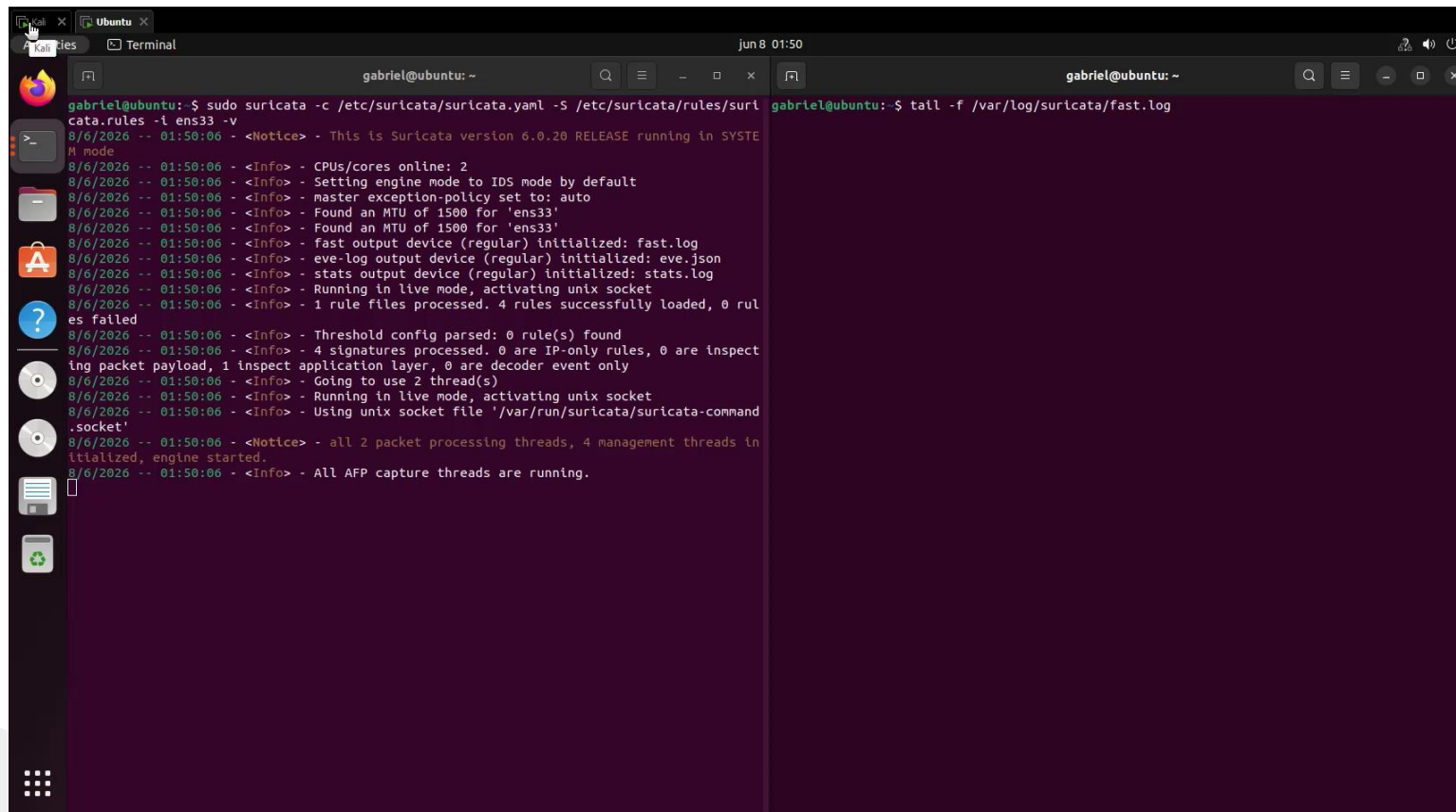
User-Agent



The screenshot shows a Kali Linux desktop environment. On the left is the application menu sidebar. In the center, a terminal window displays the output of running Suricata. On the right, a web browser window shows the same terminal output, indicating it's a live stream or a saved page. The terminal output shows Suricata version 6.0.20 running in live mode, with various configuration details and a notice that all packet processing threads are running.

```
ubuntu@ubuntu:~$ sudo suricata -c /etc/suricata/suricata.yaml -S /etc/suricata/rules/suricata.rules -i ens33 -v
8/6/2026 -- 01:25:20 - <Notice> - This is Suricata version 6.0.20 RELEASE running in SYSTEM mode
8/6/2026 -- 01:25:20 - <Info> - CPUs/cores online: 2
8/6/2026 -- 01:25:20 - <Info> - Setting engine mode to IDS mode by default
8/6/2026 -- 01:25:20 - <Info> - master exception-policy set to: auto
8/6/2026 -- 01:25:20 - <Info> - Found an MTU of 1500 for 'ens33'
8/6/2026 -- 01:25:20 - <Info> - Found an MTU of 1500 for 'ens33'
8/6/2026 -- 01:25:20 - <Info> - fast output device (regular) initialized: fast.log
8/6/2026 -- 01:25:20 - <Info> - eve-log output device (regular) initialized: eve.json
8/6/2026 -- 01:25:20 - <Info> - stats output device (regular) initialized: stats.log
8/6/2026 -- 01:25:20 - <Info> - Running in live mode, activating unix socket
8/6/2026 -- 01:25:20 - <Info> - 1 rule files processed. 2 rules successfully loaded, 0 rules failed
8/6/2026 -- 01:25:20 - <Info> - Threshold config parsed: 0 rule(s) found
8/6/2026 -- 01:25:20 - <Info> - 2 signatures processed. 0 are IP-only rules, 0 are inspecting packet payload, 0 inspect application layer, 0 are decoder event only
8/6/2026 -- 01:25:20 - <Info> - Going to use 2 thread(s)
8/6/2026 -- 01:25:20 - <Info> - Running in live mode, activating unix socket
8/6/2026 -- 01:25:20 - <Info> - Using unix socket file '/var/run/suricata/suricata-command.sock'
8/6/2026 -- 01:25:20 - <Notice> - all 2 packet processing threads, 4 management threads initialized, engine started.
8/6/2026 -- 01:25:20 - <Info> - All AFP capture threads are running.
```

Blog Exec



The screenshot shows a Kali Linux terminal window with the following content:

```
gabriel@ubuntu:~$ sudo suricata -c /etc/suricata/suricata.yaml -S /etc/suricata/rules/suricata.rules -l ens33 -v
8/6/2026 -- 01:50:06 - <Notice> - This is Suricata version 6.0.20 RELEASE running in SYSTEM mode
8/6/2026 -- 01:50:06 - <Info> - CPUs/cores online: 2
8/6/2026 -- 01:50:06 - <Info> - Setting engine mode to IDS mode by default
8/6/2026 -- 01:50:06 - <Info> - master exception-policy set to: auto
8/6/2026 -- 01:50:06 - <Info> - Found an MTU of 1500 for 'ens33'
8/6/2026 -- 01:50:06 - <Info> - Found an MTU of 1500 for 'ens33'
8/6/2026 -- 01:50:06 - <Info> - fast output device (regular) initialized: fast.log
8/6/2026 -- 01:50:06 - <Info> - eve-log output device (regular) initialized: eve.json
8/6/2026 -- 01:50:06 - <Info> - stats output device (regular) initialized: stats.log
8/6/2026 -- 01:50:06 - <Info> - Running in live mode, activating unix socket
8/6/2026 -- 01:50:06 - <Info> - 1 rule files processed. 4 rules successfully loaded, 0 rules failed
8/6/2026 -- 01:50:06 - <Info> - Threshold config parsed: 0 rule(s) found
8/6/2026 -- 01:50:06 - <Info> - 4 signatures processed. 0 are IP-only rules, 0 are inspecting packet payload, 1 inspect application layer, 0 are decoder event only
8/6/2026 -- 01:50:06 - <Info> - Going to use 2 thread(s)
8/6/2026 -- 01:50:06 - <Info> - Running in live mode, activating unix socket
8/6/2026 -- 01:50:06 - <Info> - Using unix socket file '/var/run/suricata/suricata-command.sock'
8/6/2026 -- 01:50:06 - <Notice> - all 2 packet processing threads, 4 management threads initialized, engine started.
8/6/2026 -- 01:50:06 - <Info> - All AFP capture threads are running.
```

On the right side of the terminal window, the command `tail -f /var/log/suricata/fast.log` is entered, but no output is visible yet.

Conclusão

Sistemas de Detecção de Intrusões programáveis são o novo padrão **exigido** na cibersegurança moderna.



Obrigada pela a atenção!

Artur Falcão

a2024130485@isec.pt

Gabriel Rodrigues

a2024133397@isec.pt

Rodrigo Prazeres

a2023139373@isec.pt

Referências

- [1] Open Information Security Foundation (OISF), Suricata 6.0.20 Documentation, 2026, documentação oficial da versão utilizada no laboratório. [Online]. Available: <https://docs.suricata.io/en/suricata-6.0.20/index.html>
- [2] R. Ierusalimschy, L. H. de Figueiredo, and W. Celes, Lua 5.1 Reference Manual, 2006. [Online]. Available: <https://www.lua.org/manual/5.1/>